

Midterm 2 Part c

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I. QUESTION (4 POINTS / 2 LINES)

```
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Write Python code to:

- Reshape it into a 3x3 matrix.
- Extract the middle row.

II. QUESTION (6 POINTS / 3 LINES)

```
df = pd.DataFrame({  
    "Name": ["Alice", "Bob", "Charlie"],  
    "Score": [85, 92, 88],  
    "Passed": [True, True, False]  
})
```

- Write the code to filter rows where Score > 90.
- Add a new column Grade based on Score: A for Score >= 90 B for Score >= 80 C otherwise.
- Write a Pandas code snippet to calculate the average score of students grouped by whether they passed or failed, using the DataFrame above.

III. QUESTION (6 POINTS / 6 LINES)

Write the Matplotlib code to:

- Create a bar plot showing the number of students in three classes:
- Class A (10 students), Class B (15 students), and Class C (7 students).
- Label the x-axis as "Classes" and the y-axis as "Number of Students."
- Title is "Number of Students in Each Class"

IV. QUESTION (6 POINTS / 6 LINES)

Use plt.scatter() to create a scatter plot for the points (1, 2), (2, 4), (3, 6), and (4, 8), and label the axes with "X-axis" and "Y-axis". The title is "Scatter Plot of Points".